

Layout Generator Assignment – Internship

1. Introduction

The objective of this assignment is to design a small program that automatically generates multiple valid building layouts for a given rectangular site while respecting a set of geometric and spatial rules. The solution demonstrates basic design automation using Python and visualization techniques.

2. Site Setup

- Site Size: 200 m × 140 m rectangular plot
- Tower A Footprint: 30 m × 20 m
- Tower B Footprint: 20 m × 20 m

A Cartesian coordinate system is used where the site spans from (0,0) to (200,140).

3. Placement Rules

1. All buildings must be completely inside the site boundary.
2. Minimum distance between any two buildings is 15 m (edge to edge).
3. Minimum distance from building to site boundary is 10 m.
4. For every Tower A, at least one Tower B must exist within 60 m.
5. A central 40 m × 40 m plaza is reserved where no buildings are allowed.

4. Approach Used

A random search based approach is used to explore multiple building configurations. Buildings are randomly placed within the site, and each placement is validated against the given rules. Invalid placements are rejected, and valid layouts are stored and visualized.

5. Implementation

The solution is implemented in Python using the Matplotlib library for visualization. The program generates multiple layouts, checks rule compliance, and visually represents each valid configuration.

6. Output & Results

- Multiple valid layouts are generated automatically.
- Each layout clearly shows site boundary, central plaza, and building footprints.
- Tower A buildings are shown in red and Tower B buildings in blue.
- All placement rules are satisfied in the generated layouts.

7. Conclusion

This assignment demonstrates how computational techniques can assist in early-stage architectural layout planning. The approach is simple, scalable, and can be extended with optimization or machine learning techniques for improved design exploration.

8. How to Run the Program

1. Install required library using: `pip install matplotlib`
2. Run the Python file: `python layout_generator.py`
3. The layouts will be displayed automatically.